

Shanmukha Srinivas Challa

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 4 years of hands-on Python and full-stack development experience across projects, academic work, and 1 year of professional internships. Reduced API latency by processing 1M+ records in ~3 seconds and improved deployment reliability by 40% through CI/CD automation. Proficient in Python, FastAPI, Flask, React, PostgreSQL, Docker, and AWS — with a consistent track record of building production-grade systems, clean integrations, and test-driven delivery.

EXPERIENCE

Software Engineer Intern | Virtusa, Hyderabad, India

Jan 2024 – Jun 2024

- Designed and optimized C# / ASP.NET Core REST APIs to process 1M+ records in ~3 seconds, enabling near-real-time batch analytics.
- Refactored React UI components into reusable patterns and shared utilities, reducing front-end development effort by 30%.
- Implemented CI/CD pipelines and deployment gates in Azure DevOps and Jenkins, cutting deployment time by 40% and improving release reliability.

Software Engineer Intern | Sanjana Logistics LLP, Visakhapatnam, India

May 2023 – Jul 2023

- Optimized Oracle SQL queries and indexing strategies, improving query execution speed by 45% for reporting and operational workloads.
- Built KPI dashboards using React and Spring Boot, improving decision turnaround by 15% through enhanced visibility and drill-down reporting.
- Migrated infrastructure to AWS EC2 and S3, improving scalability while reducing infrastructure costs by 20%.

PROJECTS

Full-Stack FastAPI Calculator | FastAPI, PostgreSQL, JWT, React, Alembic, Pytest, Playwright

- Built a JWT-authenticated CRUD web app with user-scoped workflows and export features (CSV/JSON/PDF), cutting manual reporting time by 50%.
- Achieved ~96% test coverage across 194 automated unit, integration, and end-to-end tests, significantly improving release confidence.

Medical Triage + RAG — Evidence-Based Q&A with Citations | Python, RAG, Embeddings, Vector Search, FastAPI

- Built a retrieval-augmented medical Q&A assistant delivering citation-grounded answers, achieving 90%+ citation coverage on evaluation set.
- Improved citation accuracy by 15% across 50 evaluation questions by optimizing ingestion, retrieval, and re-ranking pipelines.

Full-Stack Banking Web App | Flask, SQL, Python

- Developed onboarding and transaction workflows with role-based admin access and audit-friendly history, reducing manual effort by 50%.
- Added input validation and workflow automation that improved end-to-end transaction processing time by ~35% and ensured full traceability.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript/TypeScript, C/C++, C#, HTML/CSS

Frameworks: FastAPI, Flask, Spring Boot, React, Angular, Node.js/Express, Next.js, Django; Pytest, JUnit

ML / AI: Deep Learning, NLP, Transformers (BERT, GPT, CLIP), TensorFlow, PyTorch, Scikit-learn, LightGBM, RAG, Embeddings

MLOps / DevOps: Docker, Kubernetes, Jenkins, Azure DevOps, GitHub Actions, Terraform, Ansible, Packer, Tekton

Data / Cloud: ETL, Warehousing, Kafka, Pub/Sub; PostgreSQL, MySQL, Oracle, MS SQL, MongoDB; AWS, Azure, GCP

Tools / BI: Git, Postman, Splunk, JIRA; Power BI, Tableau

EDUCATION

M.S. Data Science — New Jersey Institute of Technology, Newark, NJ

GPA: 3.8 | Aug 2024 – May 2026

B.Tech. Computer Science & Engineering — Gayatri Vidya Parishad College (A), India

GPA: 3.3 | Jul 2020 – Jul 2024

ACHIEVEMENTS & CERTIFICATIONS

Awards: 3rd Place – Hack-The-Mountain Hackathon (Fraud Detection: AWS + LightGBM) | Finalist – ENCRYPT 2022: Blockchain identity verification tool, reducing verification time by 35% | IEEE CS Organizing Lead; Youth for Sustainability Volunteer; Rotaract Member

Certifications: AWS ML Foundations | AWS Cloud Foundations | AWS Data Analytics | Research Publication: Impact of IoT on Education Sector After COVID-19 (Business & Management Journal, India)